# Milestone 2 Documentation

**Project Title:** AI-Powered-Enhanced EHR Imaging & Documentation System
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## Introduction

The aim of this project is to enhance medical images using AI techniques to improve clarity and assist in diagnosis. Medical images such as MRI, CT, and X-ray scans often contain noise or low resolution, making it harder for clinicians to interpret. By applying Generative AI and classical image enhancement methods, we aim to denoise, reconstruct, and improve visualization for better diagnostic support.

This milestone focuses on implementing and testing different enhancement techniques while addressing practical challenges in preprocessing, model training, and evaluation.

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## Objective of Milestone 2

The objective of Milestone 2 is to **implement and demonstrate basic image enhancement techniques on medical images** using both classical methods and Generative AI. Specifically:

- Apply GenAI for denoising and realistic reconstruction of medical images.
- Improve resolution and clarity for better clinical visualization.
- Train and validate image enhancement models using open-source libraries and AI tools.

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## Methodology

### Step 1: Choose a Dataset

- Used the medical images collected in Milestone 1 (MRI, CT, X-ray).
- Selected ~20–50 sample images for testing enhancement.
- Dataset was small but sufficient for demonstrating enhancement methods.

**Step 2: Preprocess the Images**

- Converted all images into a standard format (.png, 256x256 pixels).
- Normalized pixel values between 0.0 and 1.0.
- Split dataset into training and testing subsets for fair evaluation.

**Challenges in Step 2:**

- Switching from **OpenCV to PIL**: PIL provided a simpler workflow for loading/saving images.
- Handling **unstructured data**: Different formats (.jpg, .png) and sizes required standardization.
- Normalizing pixel values correctly to [0,1] range.
- Creating a **reliable train-test split** while maintaining randomness.

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**Step 3: Pick a Technique for Enhancement**

Two approaches were explored:

1. **Classical Super-Resolution & Denoising Models**
    - Tried models like **DnCNN, EDSR, and SRCNN** using PyTorch/TensorFlow.
    - Low-resolution images were generated from high-resolution ones using **bicubic downsampling**.

    - Faced challenges with training speed and computational requirements on CPU.

2. **Generative AI-based Enhancement**
    - Used AI APIs (OpenAI/Gemini) to enhance medical images.
    - Input: noisy/low-res image → Output: cleaner/high-res image suitable for diagnostic visualization.

**Challenges in Step 3:**

- **DnCNN and EDSR** produced unsatisfactory or blurry results.
- **High computational cost**: Training on CPU was extremely slow → required GPU support.
- Had to maintain a proper **folder structure** (models, logs, results) for organization.
- Difficulties in **downsampling datasets** to generate synthetic low-resolution images.
- Saving outputs and logs → solved by **auto-zipping results** for sharing.

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**Step 4: Validate the Improvement**

- Compared **before vs. after images** (visual evaluation).
- Used simple quantitative metrics:
    - **PSNR (Peak Signal-to-Noise Ratio)**
    - **SSIM (Structural Similarity Index)**

**Challenges in Step 4:**

1. Lack of ground-truth high-quality images → metrics didn't always reflect true visual improvements.
2. Large number of enhanced images (~100+) made handling outputs difficult.
3. Limited **Google Colab resources** restricted long training runs.
4. Deep learning models gave **blurry outputs**, so fallback on **OpenCV enhancements** for reliability.

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## Implementation

- Used Python libraries: **OpenCV, PIL, TensorFlow, PyTorch, Matplotlib**.
- Implemented preprocessing pipelines (resizing, normalization, splitting).
- Applied both **traditional super-resolution models** and **Generative AI-based enhancement**.
- Saved input vs. output images (Enhanced vs. Blurred) for clear comparison.

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## Results

- Enhanced images showed **better clarity, reduced noise, and improved sharpness** compared to blurred/noisy inputs.

- **Visual inspection** confirmed improvements in diagnostic quality.
- Quantitative results (PSNR/SSIM) showed modest improvements but did not always match visual perception.

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## Conclusion & Next Steps

We successfully implemented multiple enhancement approaches (DnCNN, EDSR, OpenCV, GenAI APIs) and tested them on medical datasets. While deep learning models required high computational resources and sometimes gave unsatisfactory results, Generative AI and OpenCV methods provided practical improvements.

**Next Steps:**

- Automate the entire pipeline for preprocessing, enhancement, and validation.
- Use larger, more diverse medical datasets for training.
- Explore hybrid models combining **classical + generative AI** for more robust performance.
- Integrate clinician feedback to ensure enhancements are clinically meaningful.

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## References


- OpenCV Documentation
- PyTorch Image Super-Resolution Models
- TensorFlow Super-Resolution Tutorial
- Research papers on medical image denoising and super-resolution